CSE 431: ML and DL Project Work Guideline

Marks: 20

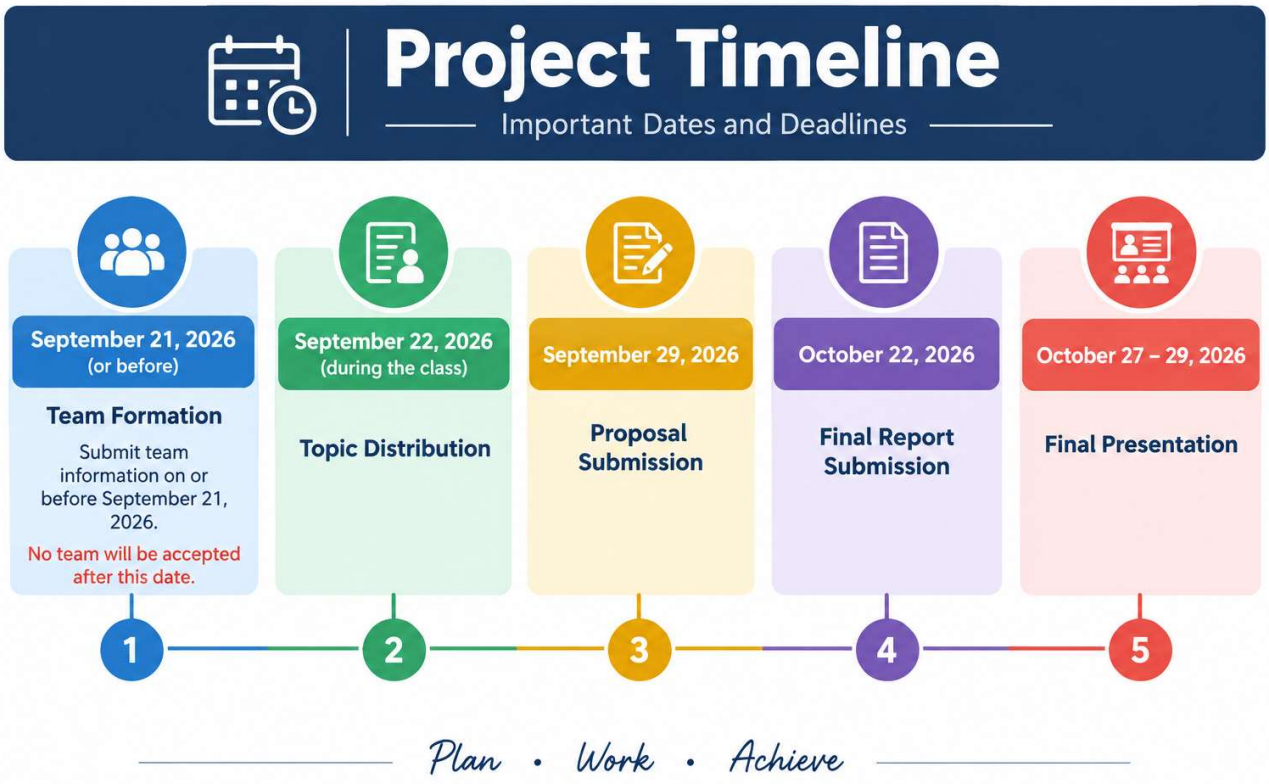
Students need to form teams to complete the project.

Team Formation Guidelines

1. **Each team must consist of exactly 3 students.**
2. **Students should form teams based on complementary skills and interests**, rather than selecting members only based on friendship or convenience.
3. Ideally, a team should include students with different strengths, for example:
    - **Member 1 – ML/DL & Research:** model development, algorithm selection, experimentation.
    - **Member 2 – Data & Implementation:** data collection, preprocessing, feature engineering, coding.
    - **Member 3 – Analysis & System Development:** evaluation, visualization, interpretation, documentation, and presentation.
4. **All three members must have a meaningful technical contribution.** One student should not be responsible for most of the coding or research work.

Important Dates:

- **Team formation:** submit team information on or before September 21, 2026. No team will be accepted after this date.
- **Topic Distribution:** September 22, 2026, during the class
- **Proposal submission:** September 29, 2026
- **Final Report Submission:** October 22, 2026
- **Final Presentation:** October 27 - 29, 2026



Guideline		Proposal (5 Marks) (minimum 1–2 pages) (Proposal should be created on Google Doc)	Final Report (10 Marks) (minimum 8–10 pages) (Final Report should be created on Google Doc)
Overview and Context	**Abstract**	The problem statement is clear and concise, understandable to non-experts.	The problem statement is clearly articulated and understandable to a non-expert audience.
	Motivation & Objectives	Students articulate why the problem is important or who benefits and state at least one qualitative or quantitative objective.	Contextual motivation is provided and qualitative and quantitative objectives are clearly stated and related to the problem.t
Related Work and Background	**Literature Review**	Students cite **at least** 4 recent (2022 – 2026) previous work relevant to their project. A short explanation of how these works relate to their project is included.	Students identify relevant prior work, preferably grouping related research lines or methodologies. Proper citation and clear connections to the problem are included.
	Background	At least one cited paper connects the evaluation metric or baseline to the problem.	Cited papers explicitly connect the evaluation metric and baseline models to the problem statement
Methodology	**Model Description**	Basic explanation of the proposed model	Clear diagrams/tables describing inputs, outputs, and structure.

		(inputs/outputs). Diagrams/tables encouraged.	Output map shapes and parameter counts included.
	Dataset	Students identify a potential data source and briefly describe how it relates to the problem.	Steps for training and evaluations are explicitly stated, including batch sampling. Dataset citation provided.
	Evaluation Metric	The metric students plan to use for assessing success is mentioned.	The chosen metric is stated, with a simple explanation of its relevance. Mathematical definitions with variables clearly labeled and tied to the problem.
	Loss Function	Mentioned briefly if applicable.	Loss and optimization described with mathematical precision.
	Experimental Depth & Iteration *	Students state at least one planned model/experiment variation beyond the baseline.	Students demonstrate multiple experimental attempts or configurations showing development beyond a single training run.
Baseline and Extensions	**Baseline Selection & Evaluation**	Students propose one or more potential baselines for comparison.	**MANDATORY:** Students justify their choice of baseline, referencing relevant papers or models. Baseline evaluations are conducted systematically.
	Implemented Extensions/ Experiments	Students describe 1–2 initial ideas for how their work might extend beyond the baseline.	High-level description of extensions/experiments to be conducted. Results show improvement over baselines or justified analysis of failures.
	Baseline		Students provide clear, original baseline outputs (plots/logs/metrics) confirming reproduction.
Results and Analysis	**Results**	Anticipated results described in general terms (e.g., expected improvement over baseline). Not mandatory.	Separate and clear results for training and validation (plots/tables). Key findings visualized.
	Error / Failure Case Analysis		At least one failure mode or error pattern is identified and discussed.
	Sensitivity / Ablation Analysis		At least one comparison demonstrating how performance changes with a design choice (e.g., model variant, hyperparameter).
Discussion		Students briefly discuss potential challenges or risks in executing the proposed project.	The team explains the significance of results, potential risks, and sensitivity of results to input changes, discussing limitations where applicable.

Future Directions		Planned work: a clear outline of next steps to be done before the final report. students identify 2-3 specific actions to move forward	Students identify areas for future exploration or improvement based on results and limitations.
Conclusion			Students effectively summarize their key findings, progress made, and how the work relates to their objectives.
Team contribution (I will use the Google doc history to assess each team member's contribution level.)		Initial roles and responsibilities. (For example: **Team Member 1:** Overview and Context, Related Work and Background, Discussion, Future Directions, Conclusion **Team Member 2:** Methodology **Team Member 3:** Baseline and Extension, Result and Analysis	Provide detailed contribution breakdown by members

- **Experimental Depth & Iteration: Example**

Model	Hidden layers	Parameters/hyper-parameters
1layerCNN	1 x {CNN, Maxpooling}	Number of kernels (16, 32, 64, 96)
2layerCNN	2 x {CNN, MaxPooling}	Length of kernel (9, 15)
3layerCNN	3 x {CNN, MaxPooling}	Learning Rate (0.001, 0.005, 0.01)
4layerCNN	4 x {CNN, MaxPooling}	Batch Size (50, 100)
8layerCNN	2xCNN, {CNN, pooling}, 2xCNN, {CNN,pooling}, CNN, {CNN,pooling}	
1layerCNN_LSTM	1 x {CNN, Maxpooling}, 1 bidirectional LSTM layer	Number of kernels (16, 32, 64, 96)
2layerCNN_LSTM	2 x {CNN, Maxpooling}, 1 bidirectional LSTM layer	Length of kernel (9, 15)
3layerCNN_LSTM	3 x {CNN, Maxpooling}, 1 bidirectional LSTM layer	Learning Rate (0.001, 0.005, 0.01)
4layerCNN_LSTM	4 x {CNN, Maxpooling}, 1 bidirectional LSTM layer	Number of LSTM units (10, 20, 30, 40)) Batch Size (50, 100)
1layerCNN_Dense	1 x {CNN, Maxpooling}, Dense layers	Number of kernels (16, 32, 64, 96)
2layerCNN_Dense	2 x {CNN, Maxpooling}, Dense layers	Length of kernel (9, 15)
3layerCNN_Dense	3 x {CNN, Maxpooling}, Dense layers	Learning Rate (0.001, 0.005, 0.01)
4layerCNN_Dense	4 x {CNN, Maxpooling}, Dense layers	Number of Dense Layers (1, 2, 3, 4) Batch Size (50, 100)

Mandatory Presentation (5 Marks): If you do not present your submitted work, you will get zero out of 20!

Duration: Total 6 min for three members

Mandatory points that should be discussed in the presentation:

1. **Description of the problem:** Describe the proposed work- What are you trying to solve, and why? Explain the motivation and context - Why does this problem matter?
2. **Discussion on Related Works:** Briefly situate the problem relative to prior work (e.g., what is known, what is missing, or what is challenging).
3. State whether your work is a re-implementation, an application of an existing method to a new dataset, or a novel approach.
4. Describe the dataset used, its size, and its relevance to the problem. Show a small number of representative examples where possible.
5. Briefly mention any assumptions, preprocessing steps, or data transformations.
6. **Approach or Methods:** Describe your overall model or approach at a high level (inputs, outputs, structure). Include a flowchart or diagram of your pipeline and/or architecture where possible.
7. Highlight the most important design choices or components.

8. Clearly state the evaluation metric and baseline used, and explain them conceptually.
9. **Results and Discussion:** Present key results or expected outcomes (ii results are still in progress). Show evidence of experimentation beyond a single run where applicable. Briefly discuss what worked, what did not, and why.
10. If applicable, mention at least one limitation, failure case, or sensitivity to design choices.
11. Conclusions: Summarize the main findings or takeaways. Briefly state what was learned and what the next steps would be.

A general note: please resist the urge to describe everything in great detail to avoid going over the time limit because you will be stopped after 6 min.

Research topics from the following list will be assigned to the teams randomly.

Topic / Research Area	Possible Project
1. Reinforcement Learning & Autonomous Systems	
	1a. Drone Control with Reinforcement Learning
	1b. Reinforcement Learning for Autonomous Robot Navigation in Dynamic Environments
2. Multimodal AI & Intelligent Communication	
	2a. Multimodal Vehicle-to-Vehicle Beam Prediction
	2b. Multimodal Emotion Recognition Using Speech, Facial Expressions, and Text
3. Large Language Models & AI Agents	
	3a. Enabling Large Language Models as Intelligent Agents
	3b. Retrieval-Augmented Generation for Domain-Specific Question Answering
4. Speech AI & Healthcare	
	4a. Detecting Neurodegenerative Disorders from Speech
	4b. Speech Emotion Recognition Using Deep Neural Networks
5. Silent Speech & Brain–Computer Interfaces	
	5a. Silent-Speech Prediction with Deep Learning Methods
	5b. Deep Learning-Based Speech Reconstruction from Non-Acoustic Signals
6. Climate & Environmental AI	
	6a. Decoding Climate Extremes: A Deep Learning Approach
	6b. Spatiotemporal Deep Learning for Air Quality Prediction
7. Deepfake Detection & AI Security	
	7a. Reinforcement Learning-Based Neural Architecture Search for Deepfake Voice Detection
	7b. Multimodal Deep Learning for Deepfake Audio-Video Detection
8. Financial AI & Time-Series Forecasting	
	8a. Deep Learning Model for Financial Forecasting
	8b. Stock Prediction Using Transformer-Based Time-Series Models
9. AI in Agriculture	

	9a. AI-Driven Crop Disease Detection and Sustainable Management
	9b. Multimodal AI for Crop Disease Diagnosis and Yield Prediction
10. AI for Cybersecurity	
	10a. Transformer-Based Detection and Classification of Cybersecurity Threats from Network Traffic
	10b. Graph Neural Network-Based Intrusion Detection in Computer Networks
11. Generative AI & AI in Education	
	11a. Generative AI for Automated Educational Content Generation and Question Answering
	11b. LLM-Based Personalized Learning and Feedback System for University Students
12. Privacy-Preserving & Low-Resource AI	
	12a. Federated Learning for Privacy-Preserving Healthcare Prediction
	12b. Few-Shot Learning for Low-Resource Bangla Text Classification
	12c. Parameter-Efficient Fine-Tuning of Language Models for Low-Resource Bangla NLP